

CLINICAL INVESTIGATION

Eye

PROTON BEAM RADIOTHERAPY FOR UVEAL MELANOMAS AT NICE TEACHING HOSPITAL: 16 YEARS' EXPERIENCE

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Purpose: To present the results of uveal melanomas treated at Nice Teaching Hospital.

Methods and Materials: This retrospective study included 886 consecutive patients referred to our clinic for the treatment of uveal melanomas by proton beam radiotherapy from June 1991 to December 2007. Survival rates were determined by using Kaplan-Meier estimates, and prognostic factors were evaluated using the log-rank test or Cox model.

Results: The number (percent total) of subjects staged according to the TNM classification system (6th edition) of malignant tumors included 39 stage T1 (4.4%), 420 stage T2 (47.40%), 409 stage T3 (46.16%), and 18 stage T4 (2.03%) patients. The median follow-up was 63.7 months. The Kaplan-Meier overall survival rate at 5 years according to the sixth edition TNM classification was 92% for T1, 89% for T2, 67% for T3, and 62% for T4; and at 10 years, 86% for T1, 78% for T2, 43% for T3, and 41% for T4. Five factors were found to be associated with an increased death rate: advanced age, tumor thickness, largest tumor basal diameter, tumor volume, and tumor volume-to-eyeball volume ratio. The metastasis-free survival rates were 88.3 % at 5 years and 76.4 % at 10 years. The local control rates were 93.9% at 5 years and 92.1% at 10 years. The ocular conservation rates were 91.1% at 5 years and 87.3% at 10 years.

Conclusions: We report the results of a large series of patients treated for uveal melanomas with a very long follow-up. Despite the large tumor volume treated, our results were similar to previously published findings relating to proton beam therapy. © 2010 Elsevier Inc.

Uveal melanoma, Proton beam radiotherapy, Survival, Local control, Enucleation, Ocular complications.

INTRODUCTION

Uveal melanomas are the main type of primary intraocular tumor in adults (1). However, their incidence in France of 0.7 cases per 100,000 people remains low (2). The management of such tumors involves enucleation or conservative treatments. The latter treatments include mainly transscleral local resection, brachytherapy, and external particle radiotherapy. The last two techniques are quite similar and do not show any significant differences in terms of overall survival and metastasis-free survival rates (3). Proton beam therapy for uveal melanomas offers optimal physical properties, delivering an accurate and homogeneous dose distribution within the target volume, with maximum preservation of normal tissue. In the 1990s, a medical cyclotron, accelerating protons to 65 MeV, was installed at the Centre Antoine Lacassagne in Nice. Treatment of patients began in June 1991. Since then, all uveal melanomas referred to the Teaching Hospital in

Nice for conservative therapy have been treated with proton therapy (4).

METHODS AND MATERIALS

Patients

We studied 886 patients who had been referred to our clinic (Department of Ophthalmology, St. Roch Hospital, Nice, France) for treatment of uveal melanoma by proton beam radiotherapy from June 1991 to December 2007, with a minimum follow-up period of 6 months. Each patient had a complete examination by our oncoophthalmologist, consisting of visual acuity, slit lamp examination of sclera and anterior segment, measurement of ocular pressure, direct ophthalmoscopy with a three-mirror Goldmann or indirect ophthalmoscopy, fundus photography, fluorescein angiography, and B-mode ultrasound. Magnetic resonance imaging of melanomas reaching the papilla was performed to detect any optic nerve and orbital invasion (5). During the first visit, patients were examined by liver ultrasonography and chest radiography for early

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